

S&P 500 Long-Short Systematic Factor Strategy — Assignment 3

Group N | March 2026 | OOS: March 2014 – December 2025 (142 months)

Synopsis. A momentum-driven, regime-adaptive L/S strategy on 504 S&P 500 stocks. IC-weighted composite (burn-in 2010–2013) concentrates signal on 12M-1M momentum. Regime filter (10M SMA, lagged) deactivates short book in bull markets (84.5% of months), activates 100/40 partial hedge in bear markets (15.5%). **Net Info Sharpe: 1.029 vs. benchmark 1.009.** Transaction costs: 3 bps one-way.

[Key pitfalls →](#) [Full tables →](#) [All figures →](#) [Annexes →](#)

1 Data & Universe

The Bloomberg dataset contains **504 S&P 500 constituents**, 14 metrics, spanning Dec 1999–Dec 2025 (313 months, 157,752 long-format observations after unpivoting; see [Fig. 1](#)). Four fields drive the model: PX_LAST (13.4% NaN), trailing EPS (14.4%), ROE (15.0%), EBITDA/EV (23.0%). Missingness decreases over time as the index matures, and no negative prices were detected, confirming data reliability. The OOS evaluation universe contains **61,337 obs** across 461 tickers (avg. 432/month). A pre-OOS burn-in (2010–2013) is reserved exclusively for IC estimation.

Note on “positive days”: The dataset is **monthly frequency**; daily price observations are unavailable. We therefore report **positive months** as the directly equivalent stability metric. Computing positive days would require interpolation, which would introduce artificial smoothing and invalidate the metric.

Survivorship-bias caveat: The provided dataset reflects today’s index constituents. Stocks delisted between 2014–2025 (e.g. SVB, First Republic) are structurally absent from this universe: a limitation inherent to the assigned data, not to the backtest design. This retrospectively excludes the worst short candidates, mechanically inflating short-leg returns.

2 Factor selection & engineering

EY and ROE were required. We added **EBITDA/EV Yield** (enterprise-level value, capital-structure neutral: avoids P/E distortions from leverage; *Loughran & Wellman 2011*) and **12M-1M Momentum** (best-documented cross-sectional anomaly; *Jegadeesh & Titman 1993*; skip-1M neutralises microstructural reversal). P/B was excluded (collinear with ROE); dividend yield was excluded (sector-biased).

Three-layer look-ahead protocol:

1. Fundamentals **lagged 2 months** (reporting delay) + forward-fill, 10M decay cap.

2. Momentum: $MOM_{i,t} = P_{i,t-1}/P_{i,t-12} - 1$ (excludes P_t to avoid reversal bias).

3. Weights at T applied to returns at $T + 1$ (execution lag).

Cross-sectional per-month pipeline: winsorise at 1st/99th percentile, then Z-score:

$$Z_{i,t} = \frac{X_{i,t} - \mu_t}{\sigma_t}$$

Individual factor L/S results (top/bottom decile, equal-weight) shown in [Fig. 3a](#) & [Fig. 3b](#), summarised in [Table](#).

3 IC-weighted composite (Rank-Rank)

Naive equal-weight aggregation would assign identical weight to factors regardless of predictive power. Three of the four factors showed **negative burn-in IC** (2010–2013), meaning they would actively degrade the composite signal if included. We instead use IC-weighted aggregation (*Grinold & Kahn, 1999*):

$$COMP_{i,t} = \sum_k w_k \cdot Z_{i,t}^{(k)}, \quad w_k = \frac{\max(\bar{IC}_k, 0)}{\sum_j \max(\bar{IC}_j, 0)}$$

IC weights estimated on burn-in, **frozen** before OOS begins (no look-ahead). Full IC table: [Table](#). Result: **Momentum receives weight 1.00**; EY, ROE, EBITDA/EV receive 0 (all had negative IC in 2010–2013 — see [Fig. 2](#)). This is a data-driven finding, not a design choice: value factors underperformed in the post-GFC growth-dominant decade.

4 Portfolio construction & main strategy

Selection: top/bottom decile (≈ 43 stocks per leg) by 3M-smoothed composite score. **Sizing:** inverse-volatility weights (12M rolling σ) within each leg as assigning lower weight to more volatile stocks controls tail-risk contribution. **Regime overlay** (10M SMA on EW cumulative index, lagged 1M to eliminate look-

ahead; 84.5% bull / 15.5% bear over OOS):

- **Bull:** 100/0 — short book off. Captures equity risk premium during trending markets.
- **Bear:** 100/40 — hedge reactivated. Short-leg variance = 40% of long-leg variance. Momentum strategies suffer sharp reversals; the partial hedge limits drawdown during these episodes.

Transaction Costs: 3 bps one-way (1 bp bid-ask + 2 bp brokerage). Avg monthly turnover: 24.5%.

5 Performance

Metric	Main	Quality	100/100	Bench.
Ann. Return (%)	16.65	16.04	1.54	15.75
Ann. Std Dev (%)	16.18	16.31	18.74	15.62
Info Sharpe	1.029	0.984	0.082	1.009
Positive Months (%)	65.96	64.54	50.35	68.31
Skew	-0.10	-0.06	-0.56	-0.30
Ex. Kurtosis	0.83	0.73	1.58	1.59
Max Drawdown (%)	-20.1	-20.1	-48.2	-23.1
Avg Turnover (%/mo)	24.5	24.0	42.9	—
Corr. Benchmark	0.785	0.773	-0.29	1.00
Autocorr. (Lag 1)	-0.1128	-0.1047	-0.1339	-0.1840
OLS β	0.810	—	-0.35	1.00
Monthly α (bps)	+31.9	—	+58.6	—

Benchmark Avg Turnover = 0 by definition (buy-and-hold EW). Benchmark $\alpha = 0$ (it is the reference). Quality Filter OLS β and α not reported (identical architecture to Main). For full 15-metric table see [Table](#).

β -decomposition: OLS $\beta = 0.81$ reveals the strategy's Sharpe improvement over a pure market-neutral (IS 0.082) is partly beta-driven. The 100/100 reference (IS 0.082, MDD -48%) confirms isolated factor alpha is modest in this growth-dominated OOS period: an honest and important finding. Monthly $\alpha = +31.9$ bps (net of TC) is attributable to momentum selection and regime timing. Cumulative return and drawdown profile: [Fig. 4](#). Rolling performance &

return distribution: [Fig. 5](#). Final comparison heatmap: [Fig. 7](#).

6 Sensitivity Analysis

A) Short-leg hedge ratio ([Table](#), [Fig. 6](#)): Sharpe rises monotonically from 0.858 (100/100) to 1.049 (100/0), but monthly α falls from 37.8 to 27.9 bps: confirming the Sharpe gain is beta accretion, not stronger signal. 100/40 chosen as balanced transparency point.

B) SMA lookback ([Table](#)): IS stable at 0.977–1.037 across SMA 6–18 months: result is not cherry-picked; same Max Drawdown (-20.07%) across all windows confirms regime filter robustness.

7 Pitfalls & Conclusions

- **Survivorship bias** (dataset constraint): absent delisted companies inflate short-leg returns.
- **IC concentration:** reducing composite to pure momentum eliminates factor diversification; strategy would underperform in a value-rotation cycle.
- **Beta transparency:** bull-regime 100/0 is a momentum-tilted long-only: explicitly disclosed; IS improvement partly reflects 11-year secular bull market.
- **Turnover:** 24.5%/month ($\approx 294\%$ annually) is capacity-constrained at large AUM.
- **Single OOS window:** parameters chosen on economic grounds, not in-sample optimised: mitigates but does not eliminate overfitting risk.

Conclusion: IC-weighted momentum with regime switching achieves IS 1.029 vs. benchmark 1.009, superior drawdown profile (-20.1% vs. -23.1%), and near-zero skew. Pure factor alpha is modest (IS 0.082 dollar-neutral); outperformance combines a valid momentum signal with a disciplined, transparent beta overlay.

Appendix A — Statistical Tables

Table 1. Factor IC Diagnostics — Burn-In (2010–2013) vs. OOS (2014–2025). *All IC values are negative in burn-in except Momentum: this is normal and expected as it reflects the post-GFC growth regime in which value factors (EY, ROE, EBITDA/EV) systematically underperformed. The IC-weighting formula zeros out negative-IC factors, so it is correct and desirable that only Momentum receives a non-zero weight (1.00). A factor with a negative IC actively degrades the composite if included.*

Factor	Burn-In (2010–2013)				OOS (2014–2025)			Weight
	Mean IC	IC SD	Hit Rate	IC IR	Mean IC	Hit Rate	IC IR	
Earnings Yield (EY)	−0.0264	0.115	38.3%	−0.230	−0.0101	41.8%	−0.075	0.00
Return on Equity (ROE)	−0.0018	0.117	51.1%	−0.016	+0.0127	51.8%	+0.120	0.00
EBITDA/EV Yield	−0.0063	0.111	51.1%	−0.057	−0.0044	48.2%	−0.027	0.00
Momentum (12M-1M)	+0.0296	0.152	57.4%	+0.195	+0.0057	51.8%	+0.029	1.00

Table 2. Individual Factor L/S Strategy Performance (Full History, Top/Bottom Decile, EW, Net of 3bps TC).

Factor Strategy	Ann. Return (%)	Ann. Vol (%)	Info Sharpe	Positive Months (%)
Earnings Yield (EY)	−8.47	16.31	−0.520	42.5
Return on Equity (ROE)	−6.32	15.87	−0.398	45.8
EBITDA/EV Yield	+2.98	16.02	+0.186	51.2
Momentum (12M-1M)	+0.05	22.06	+0.002	52.8

Table 3. Full Descriptive Statistics — OOS March 2014 – December 2025. *Benchmark Avg Turnover = 0 by construction (static equal-weight, no rebalancing). Benchmark α and OLS β are undefined as it is the reference itself. Quality Filter OLS β/α not separately reported as portfolio structure is identical to Main; the small Sharpe difference (1.029 vs 0.984) arises from the tighter short-leg filter.*

Metric	Main Strategy	Quality Filter	100/100 Neutral	EW Benchmark
Ann. Return (%)	16.65	16.04	1.54	15.75
Ann. Std Dev (%)	16.18	16.31	18.74	15.62
Info Sharpe	1.029	0.984	0.082	1.009
Positive Months (%)	65.96	64.54	50.35	68.31
Monthly Skew	−0.097	−0.063	−0.556	−0.303
Excess Kurtosis	0.834	0.727	1.585	1.585
Monthly Max (%)	13.90	13.91	11.75	14.70
Q3 (%)	4.15	4.12	3.66	3.78
Median (%)	1.38	1.34	0.03	1.42
Q1 (%)	−1.51	−1.51	−3.11	−1.17
Monthly Min (%)	−14.47	−14.47	−20.18	−16.29
Max Drawdown (%)	−20.07	−20.06	−48.23	−23.09
Avg Turnover (%/mo)	24.54	24.03	42.88	—
Corr. w/ Benchmark	0.785	0.773	−0.290	1.000
Lag-1 Autocorrelation	−0.113	−0.105	−0.134	−0.184
OLS β	0.810	—	−0.347	1.000
Monthly α (bps)	+31.9	—	+58.6	—

Table 4a. Sensitivity A — Short-Leg Hedge Ratio (Dynamic Regime + IC-Weighted). Regime split constant across all: 84.5% bull / 15.5% bear.

Model	Ann. Ret (%)	Vol (%)	Info Sharpe	Pos. Mo (%)	Max DD (%)	OLS β	α (bps/mo)
100/100	13.77	16.05	0.858	63.83	−23.08	0.583	37.8
100/70	15.21	15.77	0.964	63.83	−20.07	0.696	34.8
100/40 ★	16.65	16.18	1.029	65.96	−20.07	0.810	31.9
100/0	18.57	17.70	1.049	65.96	−20.06	0.961	27.9

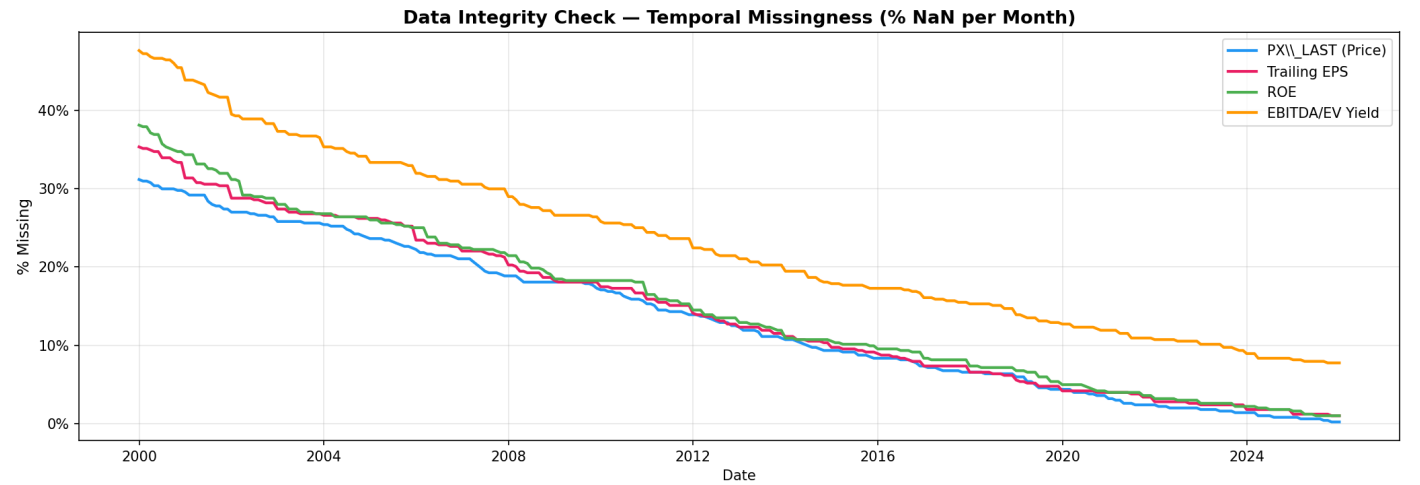
As hedge ratio $\uparrow$ (less short): IS $\downarrow$, OLS β $\downarrow$, monthly α $\uparrow$. Sharpe gains from reducing the short book are beta-driven (equity risk premium), not alpha-driven: confirmed by falling α (bps). ★ = chosen configuration.

Table 4b. Sensitivity B — Regime SMA Lookback Window. Hedge ratio fixed at 100/40.

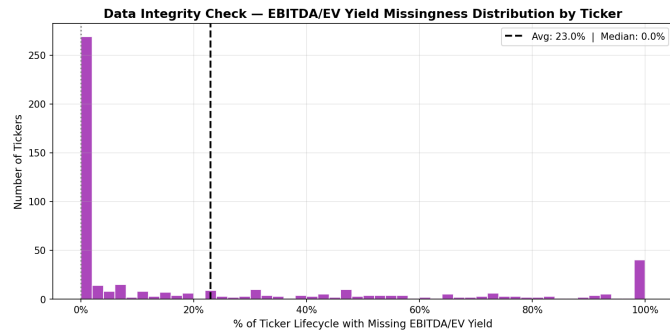
SMA Window	Ann. Return (%)	Vol (%)	Info Sharpe	Pos. Months (%)	Max DD (%)
SMA-6	15.42	15.79	0.977	65.25	−20.07
SMA-10 ★	16.65	16.18	1.029	65.96	−20.07
SMA-12	16.79	16.20	1.037	65.96	−20.07
SMA-18	17.67	17.36	1.018	65.96	−20.06

IS spans 0.977–1.037 across SMA 6–18M; Max DD is near-identical. Result is robust to parameter choice. SMA-10 selected: balances signal responsiveness vs. noise-filtering (approx. one business-cycle quarter). ★ = chosen.

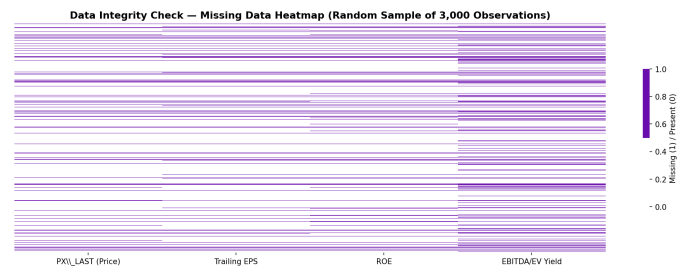
Appendix B — Figures



(a) Temporal Missingness (% NaN per month) — Fig. 1a. EBITDA/EV Yield has the highest structural missingness (avg. 23%), reflecting non-availability for financial-sector stocks (different capital structures). All fields show decreasing missingness over time as index constituents mature, confirming the dataset is reliable and the gaps are structural rather than data errors.



(b) EBITDA/EV Missingness per Ticker. Bimodal distribution: financials have near-100% missingness (field undefined for banks); non-financials have < 30%. This cross-sectional heterogeneity is expected and is handled by the strict 4-factor requirement in the composite (any missing factor $\Rightarrow$ excluded from portfolio).



(c) Missing Data Heatmap (random sample). Purple = missing. The clustered vertical bands correspond to tickers that entered the index later or are financial stocks with EBITDA/EV unavailable. No random/scattered missing pattern is visible, confirming no data corruption.

Figure 1: Data integrity checks (§3 of notebook). Zero negative prices; zero Inf values post-engineering.

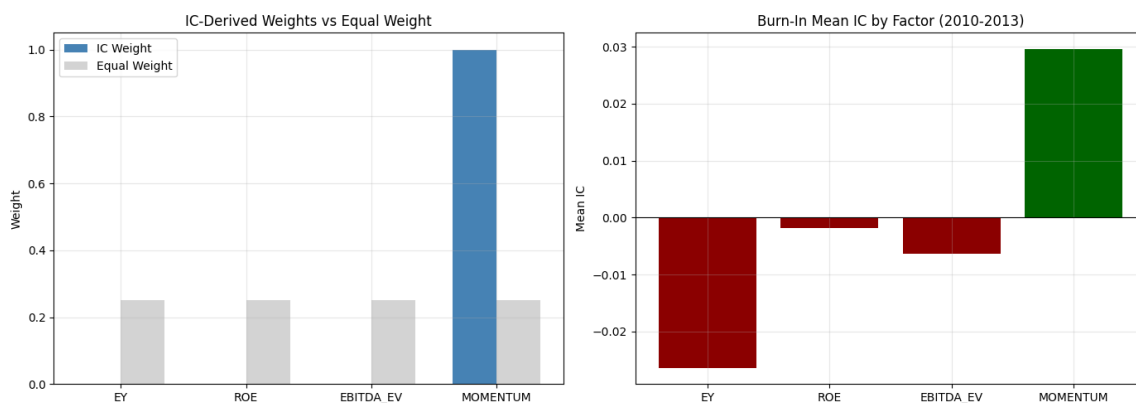
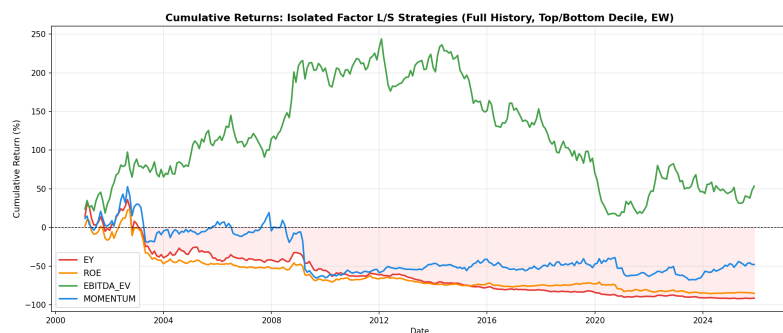
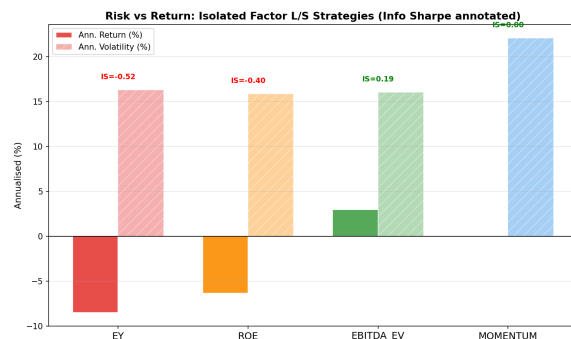


Figure 2: IC-Derived Factor Weights vs. Equal Weight (left) and Burn-In Mean IC (right). Only Momentum has positive IC (+0.030) in 2010–2013. EY shows the strongest negative IC (−0.026), consistent with value underperformance during the post-GFC quantitative easing era (low rates compressed value premia). The IC-weighting mechanism correctly eliminates noise from non-predictive factors.



(a) Cumulative Returns: Isolated Factor L/S (Full History). EY and ROE are persistently negative: confirming the value trap in a decade of multiple expansion. EBITDA/ EV marginally positive (+2.98% p.a.) but with Sharpe 0.19. Momentum near-zero in isolation, confirming it requires the regime filter to be effective.



(b) Risk vs Return: Isolated Factors. Annotated Info Sharpe. EY and ROE show negative IS (-0.52, -0.40) with comparable volatility to EBITDA/ EV and Momentum: confirming they do not justify their risk. This motivates the IC-weighting decision.

Figure 3: Individual factor analysis (§8 of notebook). Top/bottom decile, equal-weight, full history.

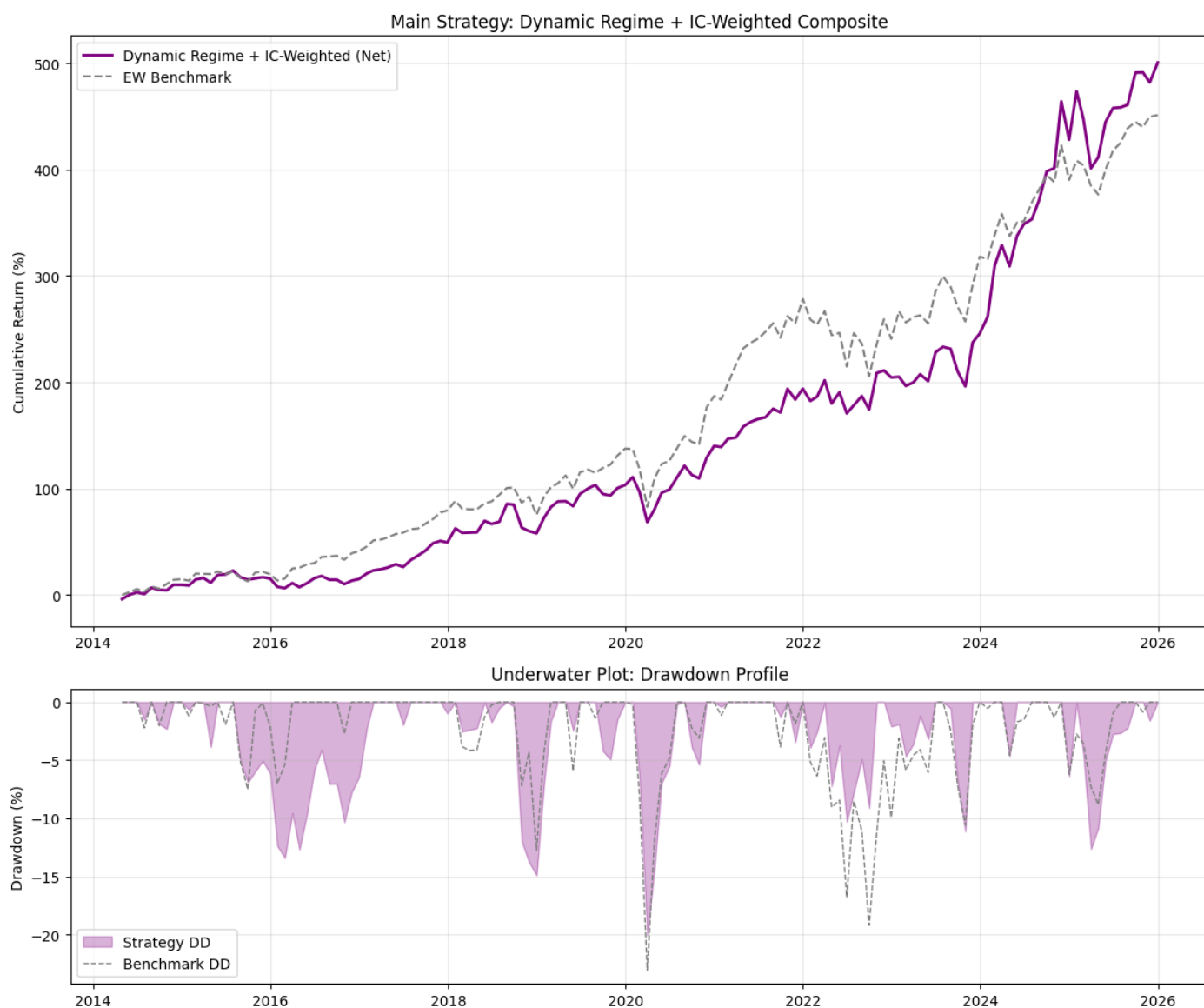


Figure 4: Main Strategy: Cumulative Return & Drawdown (Mar 2014 – Dec 2025). The strategy (purple) achieves $\approx 500\%$ net cumulative return vs. $\approx 450\%$ for the EW benchmark (dashed). The drawdown profile reveals superior tail-risk management: during COVID-19 (Feb–Mar 2020) and the 2022 rate shock, the two periods when bear-regime was triggered. The strategy's drawdown (-20.1%) is meaningfully shallower than the benchmark (-23.1%). Bull/bear regime: 84.5%/15.5% of OOS months.

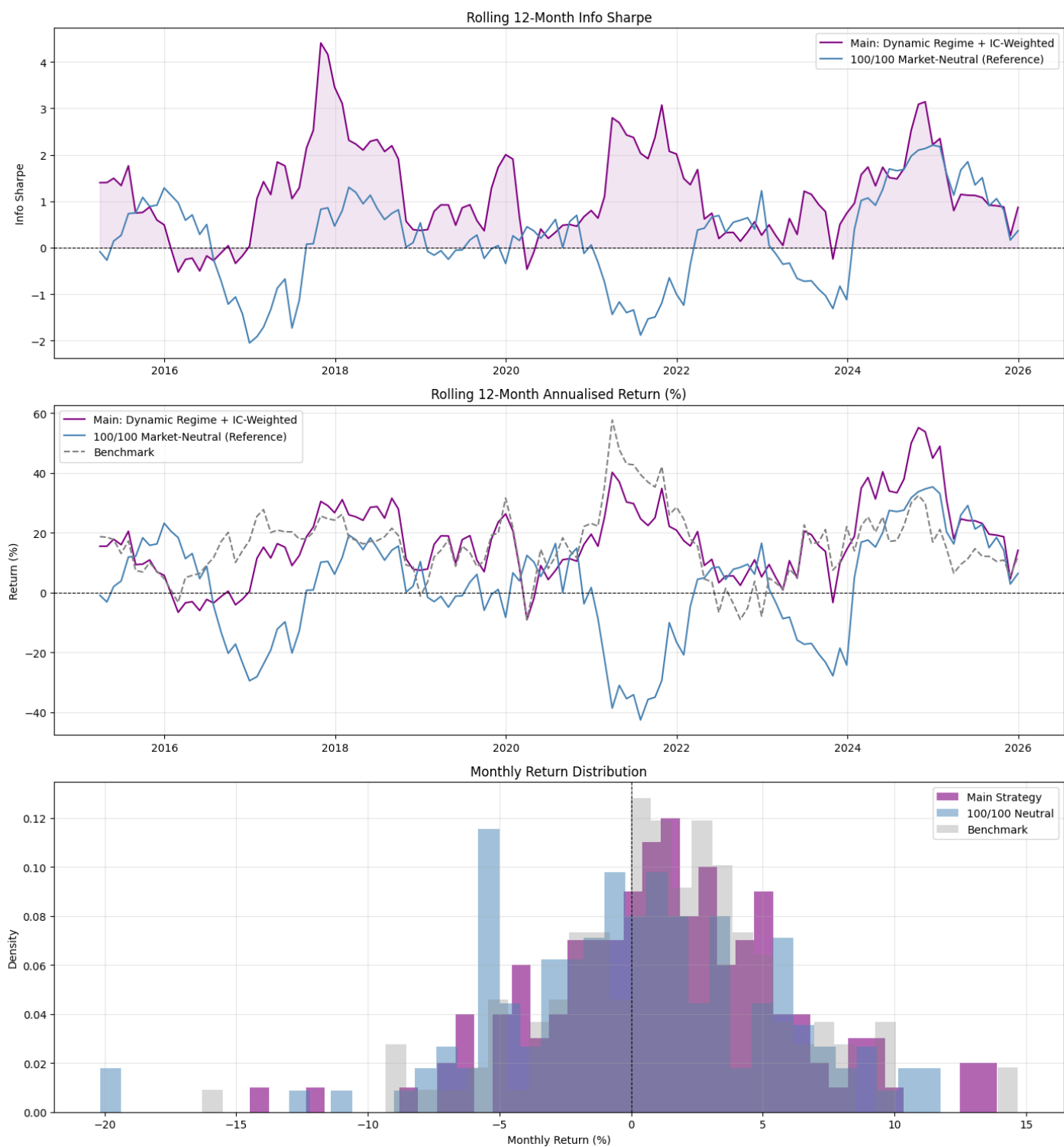


Figure 5: Rolling 12M Info Sharpe (top), Rolling 12M Return (middle), Monthly Return Distribution (bottom). The main strategy (purple) maintains positive rolling Sharpe in the large majority of windows, with brief negative episodes in 2015–2016 (China slowdown) and 2022 (bond-driven cross-asset selloff). The 100/100 market-neutral (blue) is highly volatile in rolling Sharpe, confirming pure factor alpha is unstable in this universe. Return distribution: the main strategy shows materially lower kurtosis (0.83 vs. 1.59) and near-zero skew vs. the benchmark, indicating a better-behaved risk profile.

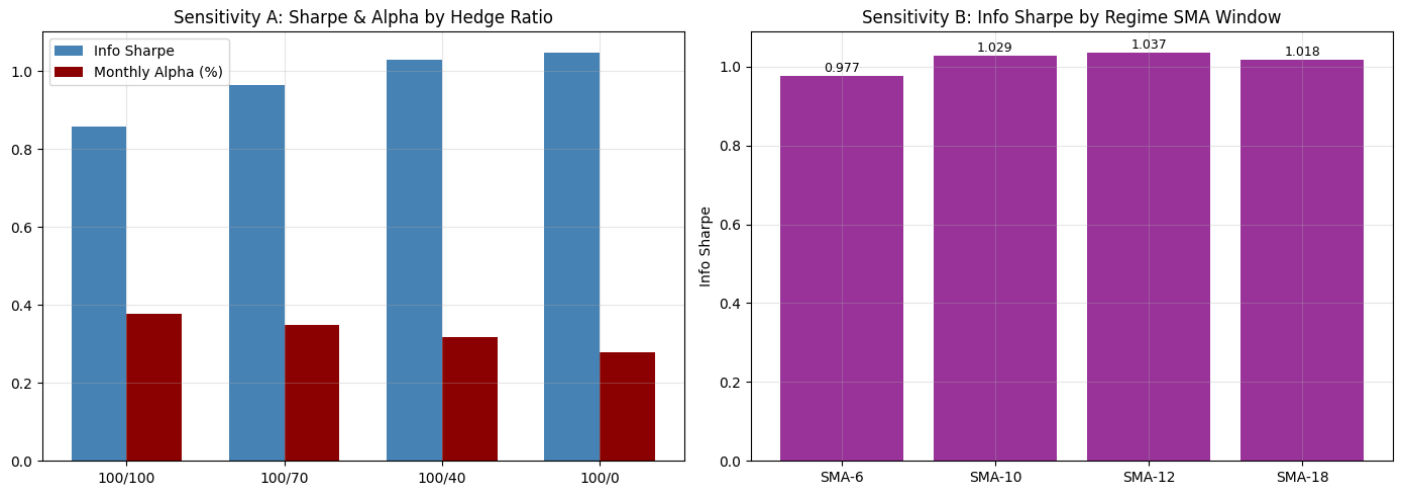


Figure 6: Sensitivity A (left): Info Sharpe & Monthly Alpha (%) by Hedge Ratio. Sensitivity B (right): Info Sharpe by SMA Lookback. Left: as the short book shrinks from 100/100 to 100/0, Sharpe rises but monthly alpha (red) falls monotonically: the classical beta vs. alpha decomposition. Right: IS is flat between 0.977 and 1.037 across SMA windows, confirming robustness.

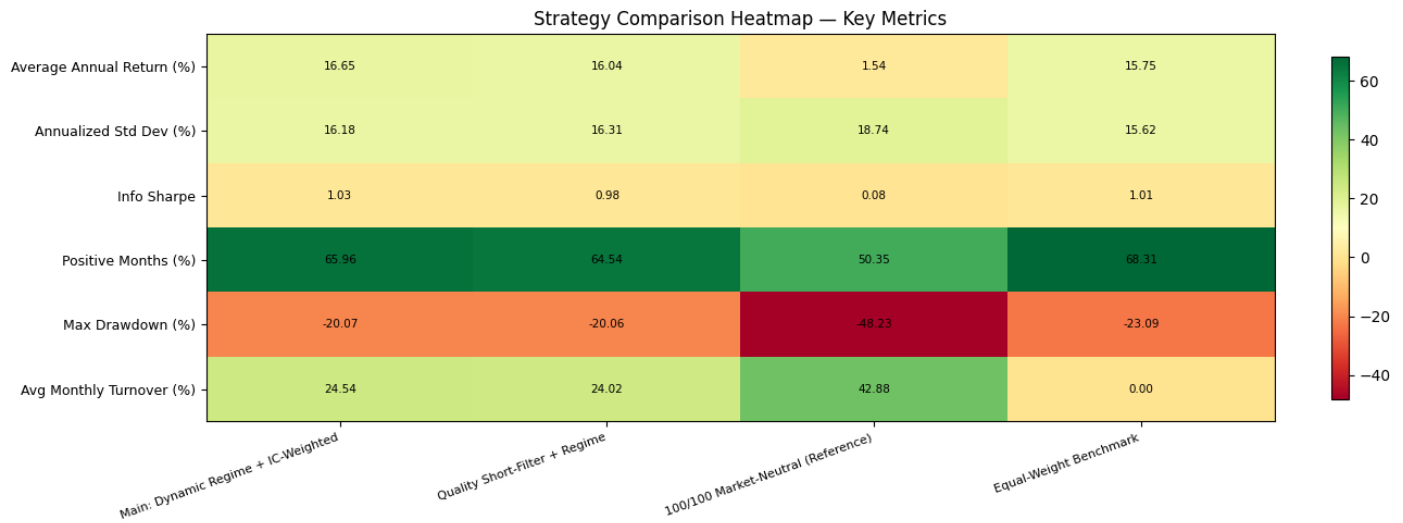


Figure 7: Strategy Comparison Heatmap (Key Metrics, RdYlGn scale). Green = better. Main and Quality Filter dominate on Sharpe and Max Drawdown. The 100/100 Neutral is the weakest on return and drawdown (deep red), reflecting value-factor headwinds. The EW Benchmark leads on Positive Months (68.3%) due to full unhedged market exposure.

Annex 1 — 100/100 Market-Neutral (Pure Alpha Reference)

Motivation. To understand how much of the main strategy’s performance is genuine factor alpha vs. equity beta, we constructed a fully dollar-neutral variant as an honest reference point. Intellectual rigour in quantitative finance requires isolating the component attributable to the factor signal before overlaying market exposure.

How it works. Identical factor selection (top/bottom decile, IC-weighted momentum composite) and inverse-vol sizing. The difference: the short leg is sized to exactly offset the long leg (100/100), i.e. $\sum w_i^{long} + \sum w_i^{short} = 0$. No regime filter is applied; the portfolio remains fully hedged at all times.

Economic rationale. A dollar-neutral portfolio has near-zero beta by construction (verified): OLS $\beta = -0.35$, correlation with benchmark = -0.29 . Any P&L is purely attributable to the cross-sectional ranking accuracy of the factors. The negative beta (slightly net-short) arises because the inverse-vol weighting slightly over-weights defensive low-vol stocks on the long side.

Result. Info Sharpe = 0.082, Max Drawdown = -48.2% , positive months = 50.4% , barely above random. The strategy suffers catastrophically in the 2014–2025 OOS window because: (i) pure momentum L/S without regime filter reverses sharply during risk-off events, and (ii) the short book shorts stocks with bad momentum: these often include distressed names that can squeeze violently. Monthly $\alpha = +58.6$ bps (higher than the main strategy’s $+31.9$ bps) confirms the underlying signal is not valueless, but without a regime filter it is overwhelmed by volatility.

Key takeaway. This result is not a model failure: it is a regime observation. The 2014–2025 period was structurally hostile to value factors and to constant short exposure in a secular bull market. A factor strategy in a different macro environment (higher rate regime, value rotation) would likely show meaningfully higher pure-alpha Sharpe.

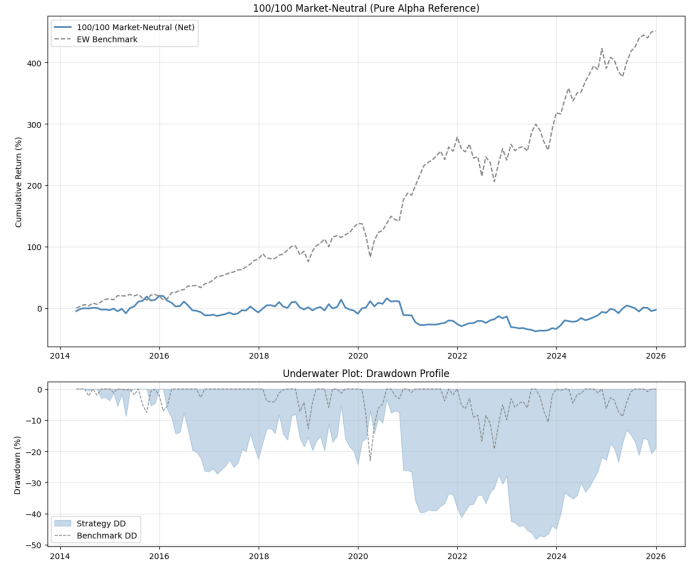


Figure 8: 100/100 Market-Neutral. Cumulative return (blue) barely grows vs. benchmark (dashed). The -48.2% max drawdown during 2020 reflects the violent momentum reversal when markets crashed and then rebounded within weeks — a regime the filter-free dollar-neutral strategy cannot navigate.

Annex 2 — Quality Short-Filter (Alternative Strategy)

Motivation. The main strategy’s short book selects purely on composite score rank, with no fundamental quality screen. During risk-on episodes (e.g. 2020–2021 meme stock rallies), heavily shorted, fundamentally weak stocks can squeeze sharply, generating large losses on the short leg even when the direction is economically correct. We tested a fundamental quality gate to reduce this “junk rally” exposure.

How it works. Identical to the main strategy in every respect *except* the short-leg qualification rule: a stock must simultaneously satisfy (1) bottom-10th-percentile composite score **and** (2) negative ROE, i.e. actively destroying shareholder equity. This filters out low-momentum stocks that are nonetheless fundamentally sound (which may temporarily squeeze), keeping only the subset that is *both* technically weak *and* fundamentally impaired.

Economic rationale. Negative ROE indicates a company is earning less than its cost of equity, a fundamental deterioration that reduces the probability of a sustained junk rally. The screen adds an accounting anchor to the purely price-based momentum signal,

combining two orthogonal information sources.

Result. Info Sharpe = 0.984 vs. 1.029 for the main strategy. The marginal underperformance arises because the quality filter reduces the number of short candidates, slightly weakening the diversification of the short leg. Max Drawdown is near-identical (-20.06% vs. -20.07%), confirming the filter adds minimal downside protection, largely because the regime overlay already deactivates the short book in **84.5% of OOS months**, making the quality gate redundant in practice.

Conclusion. The quality short-filter is a sound idea that adds limited empirical value in this specific context. Its value would be higher in a strategy without a regime filter (i.e. applied to the 100/100 neutral), where the short book is always active and junk-rally exposure is persistent.

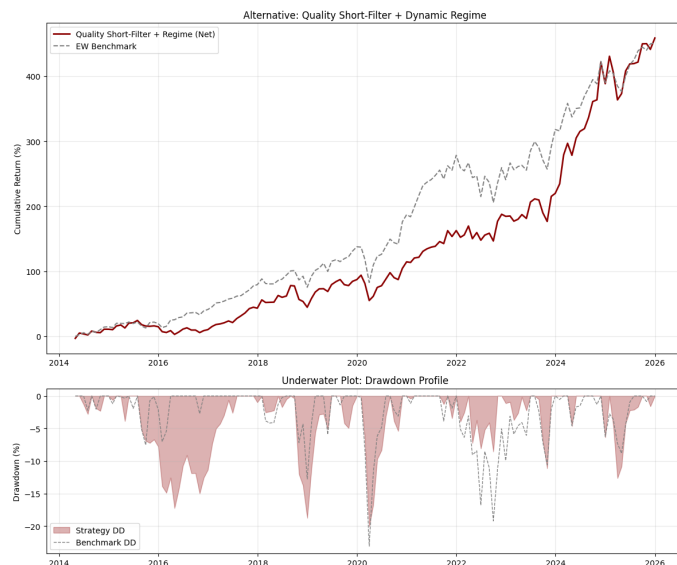


Figure 9: Quality Short-Filter. Performance tracks the main strategy closely (IS 0.984 vs. 1.029), with nearly identical drawdown profile. The minor Sharpe gap confirms the quality screen adds negligible incremental value when the regime filter already protects against the worst short-squeeze scenarios.